Index: IX_Products_ProductName

An index is like the index at the back of a book: it helps SQL Server locate matching values. This nonclustered index is on Products.ProductName. Run it only after the BEFORE measurement in PDF 08.

```
USE GreenBasketDB;
GO

CREATE INDEX IX_Products_ProductName
ON dbo.Products(ProductName);
```

Why it helps

A search such as WHERE ProductName = N'Practice product 1500' can use the name index instead of inspecting every product. SQL Server chooses the plan; a small table may still use a scan. Because the example selects only ProductID and ProductName, the index can cover the query: ProductID is the clustered primary key in this fresh schema and is included in the nonclustered index.

Cost

The index takes storage and adds work when product names are inserted, changed or deleted. Do not index every column. Measure before and after; do not claim an improvement from the index definition alone.

Check the object

```
SELECT name, type_desc
FROM sys.indexes
WHERE object_id = OBJECT_ID(N'dbo.Products');
```

Expect to see IX_Products_ProductName plus the primary-key index. The name of the automatic primary-key index is generated by SQL Server and can vary. Save the result and the actual execution plan from PDF 08.